

AURORA, OR, USA

WMO: 726959

Lat: 45.249N

Lon: 122.769W

Elev: 196

StdP: 14.59

Time Zone: -8.00 (NAP)

Period: 97-19

WBAN: 94281

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	25.4	28.2	12.1	10.3	30.8	18.5	14.1	35.1	22.9	49.9	20.3	48.5	4.5	0	0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	25.2	92.4	67.1	88.5	66.5	84.3	64.9	69.6	86.5	67.9	85.0	66.2	82.1	7.1	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.4	88.1	74.6	61.6	82.5	72.4	60.5	79.3	71.5	33.6	86.7	32.2	84.7	30.9	82.6	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	16.0	13.0	DB	20.3	100.0	4.5	3.7	17.1	102.7	14.4	104.8	11.9	106.9	8.6	109.6	(4)
(5)			WB	19.3	72.4	4.7	2.2	15.9	74.0	13.1	75.3	10.4	76.5	7.0	78.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.1	41.7	43.9	47.4	51.3	57.7	62.7	69.0	69.3	63.9	54.4	46.4	40.9		
	DBStd	11.43	6.12	5.14	4.96	5.56	6.23	5.94	5.31	5.17	5.59	5.12	5.97	6.45		
	HDD50	1043	262	177	111	46	6	0	0	0	0	18	136	287		
	HDD65	4380	721	592	546	413	241	112	18	16	86	332	557	746		
	CDD50	2551	6	6	31	85	246	382	589	599	418	153	29	7		
	CDD65	410	0	0	0	2	16	44	142	150	54	2	0	0		
	CDH74	5040	0	0	2	52	269	582	1718	1686	700	31	0	0		
	CDH80	2013	0	0	0	9	80	220	733	717	250	4	0	0		
(14) Wind	WSAvg	4.9	5.3	5.6	5.6	5.1	4.6	4.8	4.7	4.3	4.2	4.4	5.1	5.4		
(15) Precipitation	PrecAvg	42.1	6.0	3.8	5.0	3.1	2.4	1.6	0.4	0.5	1.8	3.7	6.1	7.1		
	PrecMax	55.7	13.3	6.8	9.0	6.3	4.8	3.3	1.1	1.9	7.2	10.2	14.6	14.7		
	PrecMin	28.2	2.1	0.7	1.9	1.9	1.1	0.5	0.0	0.0	0.3	0.6	2.6	1.8		
	PrecStd	7.2	3.1	1.6	1.9	1.1	1.1	0.7	0.3	0.5	1.5	2.4	2.7	3.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.9	61.5	70.3	80.9	88.0	94.3	98.6	97.4	92.5	78.9	64.4	60.6		
		MCWB	54.1	49.7	55.4	61.5	65.5	68.0	69.5	68.3	66.0	59.6	58.5	54.8		
	2%	DB	55.0	57.1	64.2	72.8	81.3	86.3	92.0	91.9	86.7	72.2	60.8	55.4		
		MCWB	51.3	49.0	52.9	57.5	62.7	65.5	67.6	66.8	64.0	57.5	55.7	52.0		
	5%	DB	52.8	54.4	60.2	66.5	75.4	81.0	88.0	88.0	82.1	67.3	57.4	52.6		
		MCWB	49.5	48.5	51.0	54.1	60.6	63.8	67.0	66.8	63.5	56.6	53.3	50.2		
	10%	DB	50.4	52.1	56.4	62.2	70.6	75.4	83.4	83.4	77.3	63.4	55.0	49.9		
		MCWB	47.6	47.6	50.0	52.5	58.1	61.0	65.4	65.3	61.5	55.8	51.5	47.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.8	54.1	57.5	62.8	67.9	70.2	73.1	71.5	69.3	63.8	61.2	56.2		
		MCDB	58.2	56.5	66.0	76.5	83.7	88.5	91.6	91.5	85.2	70.2	62.7	59.2		
	2%	WB	52.7	51.7	54.7	58.8	64.4	66.8	69.7	69.4	66.6	60.4	56.9	53.2		
		MCDB	54.2	54.6	60.3	70.4	77.2	83.5	85.5	86.8	81.6	67.2	59.7	54.9		
	5%	WB	50.0	49.9	52.7	55.7	61.7	64.4	67.7	67.7	64.6	58.5	54.3	50.5		
		MCDB	52.0	53.1	57.9	64.5	73.3	78.7	85.4	84.6	77.7	64.1	56.5	52.5		
	10%	WB	47.9	47.9	50.8	53.3	59.2	62.1	65.9	65.9	62.7	57.0	52.2	47.5		
		MCDB	50.0	51.3	55.8	60.0	69.4	73.8	82.4	81.3	74.3	62.1	54.4	49.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	13.3	15.4	17.5	19.1	20.2	25.1	25.2	23.1	17.0	12.4	10.3		
		MCDBR	12.6	17.3	23.0	28.1	29.6	30.1	32.9	33.0	32.3	25.5	14.2	11.7		
		MCWBR	9.0	10.8	12.9	14.4	14.4	13.3	12.7	12.4	13.4	13.1	9.4	9.0		
	5% WB	MCDBR	11.2	13.4	18.7	24.1	25.8	27.2	30.5	30.2	27.8	18.5	11.4	10.8		
		MCWBR	8.9	9.3	11.6	13.3	13.4	12.8	13.0	12.4	12.6	11.0	8.4	9.0		
(40) Clear-Sky Solar Irradiance	taub		0.317	0.315	0.324	0.353	0.368	0.356	0.346	0.351	0.339	0.332	0.317	0.320		
	taud		2.538	2.554	2.535	2.419	2.394	2.441	2.493	2.497	2.511	2.542	2.549	2.505		
	Ebn at Noon		254	279	290	288	285	289	290	285	280	267	252	239		
	Edh at Noon		21	24	28	35	36	35	33	32	29	24	20	20		
(44) All-Sky Solar Radiation	RadAvg		381	645	937	1362	1734	1928	2203	1882	1404	816	441	310		
	RadStd		62	91	124	126	172	177	148	122	121	99	60	48		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+266	+86
(47) Regional (26 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+157	+47

Nomenclature: See separate page